FEDERAL INSTITUTE OF SCIENCE AND TECHNOLOGY (FISAT)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SEMESTER VII

SEMINAR (CSQ413)

ACADEMIC YEAR : 2026 – 2027 **BATCH :** CSD

Seminar Title :	**Advanced Smart Contract Vulnerability Detection via LLM-Powered Multi-Agent Systems**

Student Name :	**Gautham Krishna (Roll No: 28, Slot No: 8)**

Type of Seminar :

[✓] **Review Seminar** (presenting a journal/conference/survey paper)

[ ] Technical / Concept Seminar (explaining an algorithm, technology, architecture, protocol, etc.)

[ ] Application Seminar (showing how a technology is applied in real-world systems)

[ ] Product / Tool Seminar (overview of a software/hardware product, platform, or framework)

Project Mapping

PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
x	x	x	x	x	x	x	x	x	x	x	x	x

Mapping Justification

S No	PO/PSO	Justification
1	PO1	Apply computer science and engineering knowledge to understand blockchain execution environments, smart contract semantics, abstract syntax trees (AST), and large language model architectures.
2	PO2	Analyze the complex security challenges in smart contracts—such as Reentrancy, Unsafe Delegatecall, Integer Overflow, and complex business logic flaws—and evaluate why conventional static analysis suffers from excessive false alarms.
3	PO3	Understand the design of the LLM-SmartAudit multi-agent framework incorporating role specialization (Auditor, Programming Expert, Counselor) and conversational consensus to identify contract vulnerabilities systematically.
4	PO4	Examine the paper's experimental methodology, benchmark dataset construction (110 common vulnerability contracts, 6,454 real-world Code4rena contracts, and 13 critical CVEs), and evaluation metrics (Precision, Recall, F1-score).

S No	PO/PSO	Justification
5	PO5	Understand the use of Python 3.12, multi-agent conversational frameworks, Solidity compilers (solc), baseline tools (Slither, Mythril), and frontier LLMs (GPT-4o, DeepSeek-r1) in the proposed auditing framework.
6	PO6	Recognize how securing decentralized finance (DeFi) infrastructure prevents catastrophic financial exploitation, protecting societal assets and fostering trusted digital economies.
7	PO7	Understand the critical importance of responsible disclosure, preventing dual-use weaponization of automated vulnerability scanners, and ensuring high-precision defensive auditing.
8	PO8	Develop the ability to study a peer-reviewed research paper independently from IEEE Transactions on Software Engineering, organize technical information, and collaborate in preparing and presenting the seminar.
9	PO9	Improve technical communication by explaining the multi-agent conversational architecture, Targeted vs. Broad Analysis modes, Buffer-of-Thoughts prompting, and empirical results through the presentation and seminar report.
10	PO10	Understand operational cost budgeting (~$0.21 - $0.98 USD per contract audit), token utilization management, and computational resource planning when deploying LLM-based verification pipelines in practice.
11	PO11	Develop the ability to learn emerging techniques in autonomous agent systems, dynamic smart contract fuzzing, formal verification integration, and continuous on-chain security monitoring.
12	PSO1	Apply knowledge of software design, program analysis, and AI systems to comprehend how multi-agent collaboration and prompt engineering automate vulnerability detection in Solidity smart contracts.
13	PSO2	Understand the integration of specialized LLM personas, conversational consensus algorithms, and automated audit report generation for secure, production-grade decentralized software engineering.

Signature (Seminar Coordinator) : _____